ZZ KB Nested Swings V2

Strategy & Code Logic · TradingView indicator (Pine Script v6)

When a stock falls sharply and then starts to recover, this indicator automatically maps out a **ladder of trades that ride the bounce back up**. From a single significant low (**“B”**) it finds up to four prior peaks of the decline (**T1** nearest → **T4** highest) and draws a Fibonacci entry / target / stop for each. As price climbs off the low, the trades activate one after another, like a relay.

1. The trading strategy

The idea: after a big decline, the prior highs of that decline tend to act as natural upside targets. The plan is to **buy the recovery near the bottom** and **take profit at those prior resistance levels** — repeatedly, as price works higher.

The building blocks

- **B (the common low)** — the bottom of the most recent significant decline. Everything is measured upward from here.
- **The peaks (T1…T4)** — the descending “lower highs” of the downtrend that led into B. They are numbered **nearest-to-B = T1**, up to the highest = T4. Each peak is its own trade target.
- **The Fibonacci ladder** — for each peak, levels are placed between B (0%) and that peak (100%):

Level (fraction of B→peak)	Role
`0.236`	Stop-loss line
`0.32 – 0.382`	Entry zone (the buy zone)
`0.618 – 0.68`	Golden take-profit zone
`1.0`	The peak itself (the swing high)

The rules for every trade (identical for T1–T4)

- **Entry** — a candle *closes* above the `0.382` level, **and** price is above both the 9- and 21-period EMAs, **and** volume is more than 1.2× the 20-bar average. (Three confirmations together.)
- **Take-profit** — price’s high reaches the golden zone (first touch of `0.618`).
- **Stop-loss** — a candle *closes* below the `0.236` level.

The “relay” effect: because all trades share the same low B, each trade’s golden take-profit zone sits right around the *next* trade’s entry zone. So as price recovers, the trades hand off naturally: **T1** → **T2** → **T3** → **T4**.

2. How it finds the trades (detection logic)

It uses **two independent swing detectors** (built on TradingView's ZigZag tool):

- **Decline detector** — *Min Decline %* (default 35%). Finds the significant fall and sets **B**. It uses only **confirmed** swing points, so it waits for a bottom to prove itself before marking it — it does not try to "catch a falling knife."
- **Peak detector** — *Peak Sensitivity %* (default 25%). A finer detector that locates the nested peaks above B.

The "nearest-first staircase"

Starting at B, the indicator walks **backwards in time** and keeps each peak that is meaningfully higher than the last one kept. This guarantees **T1 is the nearest peak** (not the biggest), so no in-between trade is ever hidden. Two rules keep it clean:

- **Minimum gap** — each higher peak must be at least a set % above the previous (default 8%). If two peaks are closer than that, the **higher one is kept**.
- **Stop at a lower low** — the walk halts the instant it crosses a low that is *below B*, so it can never reach into a different, older, deeper decline.

Expiry: if price has already climbed above the highest peak, the whole setup is "fully played" (nothing left to buy) and it is hidden from the chart.

3. What you see on the chart (display logic)

- Only the **active** trade (the lowest one not yet taken-profit) is drawn boldly: a golden entry zone, a green target zone, a red stop line, and labels.
- Higher, not-yet-reached trades are shown **faintly**, just for context.
- Completed and expired trades are **hidden**, to keep the chart clean.
- **Unified colours** — every trade uses the same scheme (golden / green / red), consistent with the B marker. No per-trade colour-coding.
- An optional **info table** lists every trade's levels, the size of each decline (*Depth %*), the active trade's *Risk:Reward*, and whether the filters currently pass.
- **Markers** print on the candle where each entry / take-profit / stop fires, and TradingView **alerts** can be set on each.

4. The settings (parameters)

Setting	Default	What it controls
Min Decline %	`35`	How deep a fall must be to count as a setup (this sets B)
Peak Sensitivity %	`25`	How finely the nested peaks are detected
Min Bars Between Pivots	`10`	Minimum spacing between swing points
Max Nested Swings	`4`	How many trades to stack (T1…T4)
Min Gap Between Peaks %	`8`	Minimum spacing between peaks; keeps the higher of two close ones
Fibonacci levels	`0.236 / 0.32 / 0.382 / 0.618 / 0.68`	The entry / target / stop ladder
EMA & Volume filters	`on`	Confirmation filters required for an entry
Colours & transparency	`gold / green / red`	Visual styling (unified across all trades)

5. How the code is built (technical summary)

- **Platform:** Pine Script v6, running as an overlay indicator on TradingView.
- **Two ZigZag instances** provide the two swing detectors above (one at the Min Decline %, one at the finer Peak Sensitivity %).
- **Persistent variables** remember B, the four peaks, and each trade's state across bars. Each trade is a tiny four-step state machine: `none → waiting for entry → in trade → target hit`.
- **Helper functions** keep it organised: one computes the Fibonacci levels for a swing, one runs a single trade's state machine on each confirmed bar, one draws a swing's zones.
- **All drawing is refreshed on the latest bar** (old shapes deleted, new ones drawn), so the chart never accumulates clutter.
- **The trade math is identical for every swing** — only *which* swings get detected and *how* they are displayed changed between versions. The entry / target / stop rules are unchanged.
